EXPERIMENT 13: LOW-CODE LIMITATIONS – CASE STUDY

Project: Hotel Booking Using Low-Code in Zoho Creator

AIM

To evaluate the benefits and limitations of using a low-code platform for developing a Hotel Booking application.

REQUIREMENTS

- Hotel Booking use case
- List of core requirements and future growth needs
- Zoho Creator forms, reports, workflows, and custom functions

DESIGN / MODULES

Requirement	Low-Code Fit / Consideration
Room and Booking Forms	Suitable for rapid form-based development
Status Notifications	Supported through configured workflow automation, subject to service limits
Complex Pricing Rules	May require custom functions and extensive testing
High-Volume Concurrent Bookings	Requires availability validation, capacity checks, and platform-limit review
External Payment Gateway	Requires secure integration, compliance review, and error handling

PROCEDURE

1. Define the hotel workflow: room listing, booking request, confirmation, check-in, and cancellation.
2. Map requirements to Zoho Creator forms, reports, workflows, and custom functions.
3. Identify dependencies such as external services, subscription limits, API limits, and permissions.
4. Assess risks such as double booking, inaccurate availability, data exposure, and service failure.
5. Recommend controls such as availability validation, role-based access, activity logs, and manual review for exceptions.
6. Compare the benefits and limitations of low-code development.

7. Identify requirements that may need additional engineering or specialist review.

SAMPLE LOGIC / OUTPUT

Case Study Analysis:

Feature	Evaluation
Forms and Reports	Easy to develop and maintain
Workflow Automation	Reduces repetitive manual work
Custom Pricing	Requires additional logic and testing
Concurrent Bookings	Needs reliable availability and conflict prevention
Payment Integration	Requires secure APIs and failure handling
Scalability	Must be evaluated against platform limits and expected traffic

Sample Availability Check:

```
availableRooms = Rooms[Room_Status == "Available"];


if(availableRooms.count() > 0)
{
   info "Room may be available for booking";
}
else
{
   alert "No rooms are currently available";
}
```

Note: This is illustrative logic only. A real application must check availability for the requested dates and prevent conflicting bookings. A simple status check alone cannot guarantee prevention of double booking.

TEST CASES

Scenario	Discussion Point
Two guests request the same room dates	Use a tested availability-check and conflict-prevention design

External service becomes unavailable	Display a clear status and provide safe retry or manual review
Requirement exceeds platform limits	Evaluate supported extensions or an alternative architecture
Complex pricing is calculated	Verify all pricing rules and edge cases
Unauthorized user accesses guest data	Apply and test role-based access controls

RESULT

The case study shows that low-code development in Zoho Creator can accelerate the creation of standard forms, reports, and workflow automation. However, payment processing, concurrent bookings, complex pricing, scalability, and large-scale integrations require additional engineering, security checks, and testing.